Name: ______________ Cohort/Batch: ______________ Date: ______________ Score: ______________

# CIRCLECI PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A DevOps

TOTAL: 10 QUESTIONS

**Q1** What problem does CircleCI solve in DevOps or infrastructure? — Threat Model

**Q2** What is CircleCI's core architecture? — Architecture

**Q3** How does CircleCI define infrastructure or configuration as code? — Configuration

**Q4** How does CircleCI ensure idempotency? — Hardening

**Q5** How do you handle secrets in CircleCI? — SSO / IdP

**Q6** How does CircleCI integrate with CI/CD pipelines? — Observability

**Q7** How do you debug a failed CircleCI deployment? — Lifecycle

**Q8** What are security best practices for CircleCI? — Compliance

**Q9** How does CircleCI scale for large environments? — Testing

**Q10** What are common mistakes teams make when adopting CircleCI? — Tuning

## ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

### A1 CircleCI addresses a specific concern  provisioning,

Mitigation

CircleCI addresses a specific concern  provisioning, configuration management, orchestration, or CI/CD  by automating manual steps that are error-prone, slow, or not reproducible when done by hand.

### A2 CircleCI has a control plane that stores

Primitives

CircleCI has a control plane that stores desired state and a data plane (agents or push-based SSH) that enforces it. Understanding this distinction clarifies where configuration is evaluated and where changes are applied.

### A3 CircleCI uses declarative files (YAML, HCL, or

Hardening

CircleCI uses declarative files (YAML, HCL, or a DSL) to express desired state. A plan/diff phase shows what will change before applying. Version-controlling these files enables peer review and rollback.

### A4 CircleCI operations are designed to produce the

Gotchas

CircleCI operations are designed to produce the same result when run multiple times. Applying the same config twice converges to the desired state rather than creating duplicate resources. This makes automation safe to re-run.

### A5 Secrets should never be stored in plain

Federation

Secrets should never be stored in plain text in CircleCI config files. Use a secrets backend (Vault, AWS Secrets Manager) and inject values at runtime via environment variables or encrypted vaults supported by the tool.

### A6 CI/CD pipelines call CircleCI in automated steps:

Observability

CI/CD pipelines call CircleCI in automated steps: plan on a pull request to preview changes and apply on merge to main. Approval gates can require a human to review the plan before changes go to production.

### A7 Check CircleCI's logs for the specific error

Automation

Check CircleCI's logs for the specific error message and the resource that failed. For infrastructure tools, re-run with increased verbosity (-v, --debug). Review the state file or event history to understand what changed before the failure.

### A8 Rotate credentials used by CircleCI, restrict network

Governance

Rotate credentials used by CircleCI, restrict network access to its control plane, enable audit logging of all operations, and use least-privilege service accounts. Never run CircleCI with admin credentials in production.

### A9 Large CircleCI deployments use workspaces or namespaces

Assessment

Large CircleCI deployments use workspaces or namespaces to isolate environments, remote state backends for team collaboration, and modular code to avoid a single monolithic config that is slow to plan/apply.

### A10 Common pitfalls include storing secrets in plain

Optimization

Common pitfalls include storing secrets in plain text config, skipping the plan step before apply, not reviewing state drift, and not having a rollback strategy when a deployment fails partway through.